MASTER 2 - ADEO | CY TECH 2025

# CAR PRICE PREDICTION USING MACHINE LEARNING

## Project Presentation

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# Contents

# Project Overview



- Predict the price of used cars using a dataset from Kaggle
- Dataset contains 205 cars and 27 features (technical + categorical)
- Target variable: price (continuous)
- Task type: Regression

| | car_ID | symboling | CarName | fueltype | aspiration | doornumber | carbody | drivewheel | enginelocation | wheelbase | ... | enginesize | fuelsystem | boreratio | stroke | compressionratio | horsepower | peakrpm | citympg | highwaympg | price |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 3 | alfa-romero giulia | gas | std | two | convertible | rwd | front | 88.6 | ... | 130 | mpfi | 3.47 | 2.68 | 9.0 | 111 | 5000 | 21 | 27 | 13495.0 |
| 1 | 2 | 3 | alfa-romero stelvio | gas | std | two | convertible | rwd | front | 88.6 | ... | 130 | mpfi | 3.47 | 2.68 | 9.0 | 111 | 5000 | 21 | 27 | 16500.0 |
| 2 | 3 | 1 | alfa-romero Quadrifoglio | gas | std | two | hatchback | rwd | front | 94.5 | ... | 152 | mpfi | 2.68 | 3.47 | 9.0 | 154 | 5000 | 19 | 26 | 16500.0 |
| 3 | 4 | 2 | audi 100 ls | gas | std | four | sedan | fwd | front | 99.8 | ... | 109 | mpfi | 3.19 | 3.40 | 10.0 | 102 | 5500 | 24 | 30 | 13950.0 |
| 4 | 5 | 2 | audi 100ls | gas | std | four | sedan | 4wd | front | 99.4 | ... | 136 | mpfi | 3.19 | 3.40 | 8.0 | 115 | 5500 | 18 | 22 | 17450.0 |

# Project Folder Structure



- data/raw : original dataset from Kaggle
- data/processed : cleaned dataset used for modeling
- notebooks : all Jupyter notebooks for EDA, baseline, and optimization
- models : saved trained models
- docs : data dictionary and project description
- reports : figures and presentation material
- src : optional scripts and helpers

| | | |
|---|---|---|
| Data | 29/10/2025 13:28 | Dossier de fichiers |
| Docs | 29/10/2025 13:37 | Dossier de fichiers |
| Models | 09/11/2025 21:48 | Dossier de fichiers |
| Notebooks | 10/11/2025 12:42 | Dossier de fichiers |
| Reports | 09/11/2025 22:01 | Dossier de fichiers |
| src | 29/10/2025 13:29 | Dossier de fichiers |
| READ ME | 10/11/2025 14:15 | Document texte |
| Requirements | 10/11/2025 14:19 | Document texte |

# NOTEBOOK 01: DATA OVERVIEW



- Loaded the raw CSV from data/raw
- Checked dataset shape: 205 rows, 26 columns
- Inspected first rows (df.head)
- Checked data types (numeric vs categorical)
- Checked for missing values
- Checked for duplicates
- Generated basic statistical summary

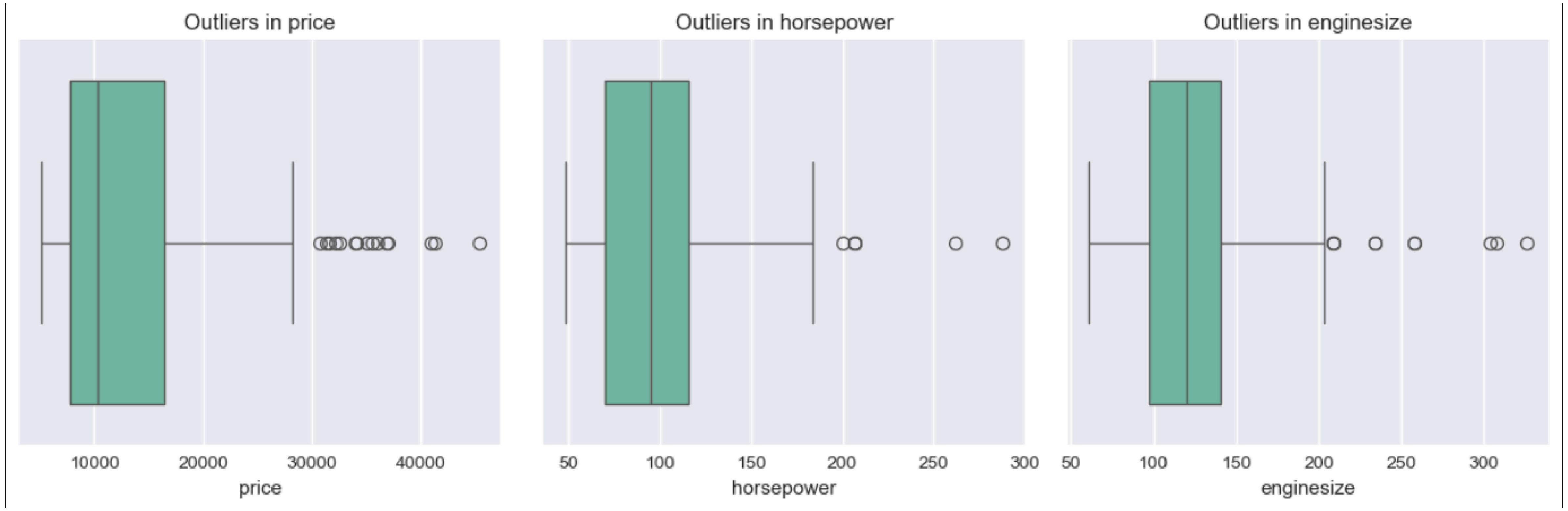
```
 Dataset info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 205 entries, 0 to 204
Data columns (total 26 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   car_ID            205 non-null    int64
 1   symboling         205 non-null    int64
 2   CarName           205 non-null    object
 3   fueltype          205 non-null    object
 4   aspiration        205 non-null    object
 5   doornumber        205 non-null    object
 6   carbody           205 non-null    object
 7   drivewheel        205 non-null    object
 8   enginelocation    205 non-null    object
 9   wheelbase         205 non-null    float64
 10  carlength         205 non-null    float64
 11  carwidth          205 non-null    float64
 12  carheight         205 non-null    float64
 13  curbweight        205 non-null    int64
 14  enginetype        205 non-null    object
 15  cylindernumber    205 non-null    object
 16  enginesize        205 non-null    int64
 17  fuelsystem        205 non-null    object
 18  boreratio         205 non-null    float64
 19  stroke            205 non-null    float64
 20  compressionratio  205 non-null    float64
 21  horsepower        205 non-null    int64
 22  peakrpm           205 non-null    int64
 23  citympg           205 non-null    int64
 24  highwaympg        205 non-null    int64
 25  price             205 non-null    float64
dtypes: float64(8), int64(8), object(10)
memory usage: 41.8+ KB
```

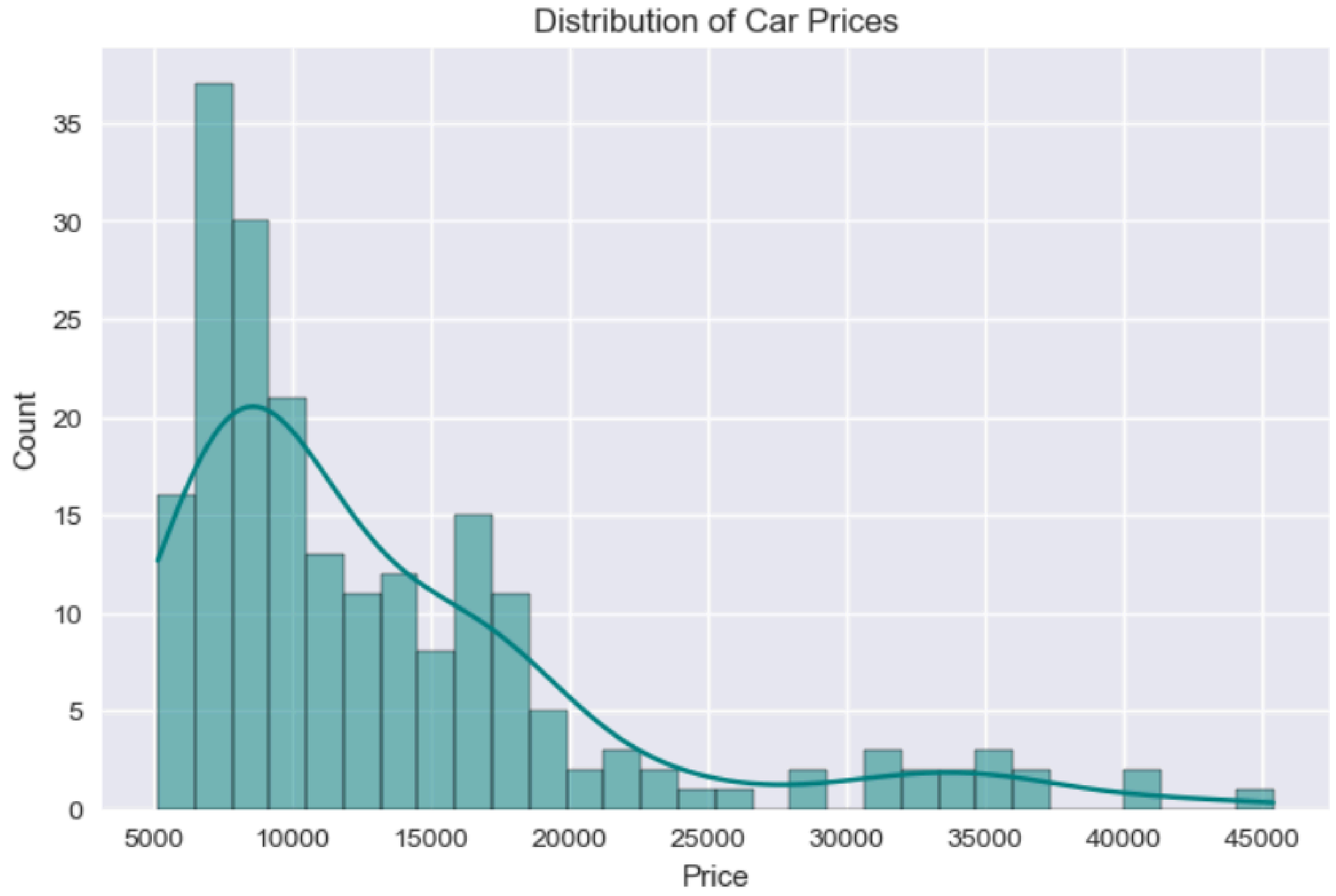
```
 Missing values in each column:
car_ID              0
symboling           0
CarName             0
fueltype            0
aspiration          0
doornumber          0
carbody             0
drivewheel          0
enginelocation      0
wheelbase           0
carlength           0
carwidth            0
carheight           0
curbweight          0
enginetype          0
cylindernumber      0
enginesize          0
fuelsystem          0
boreratio           0
stroke              0
compressionratio    0
horsepower          0
peakrpm             0
citympg             0
highwaympg          0
price               0
dtype: int64
```

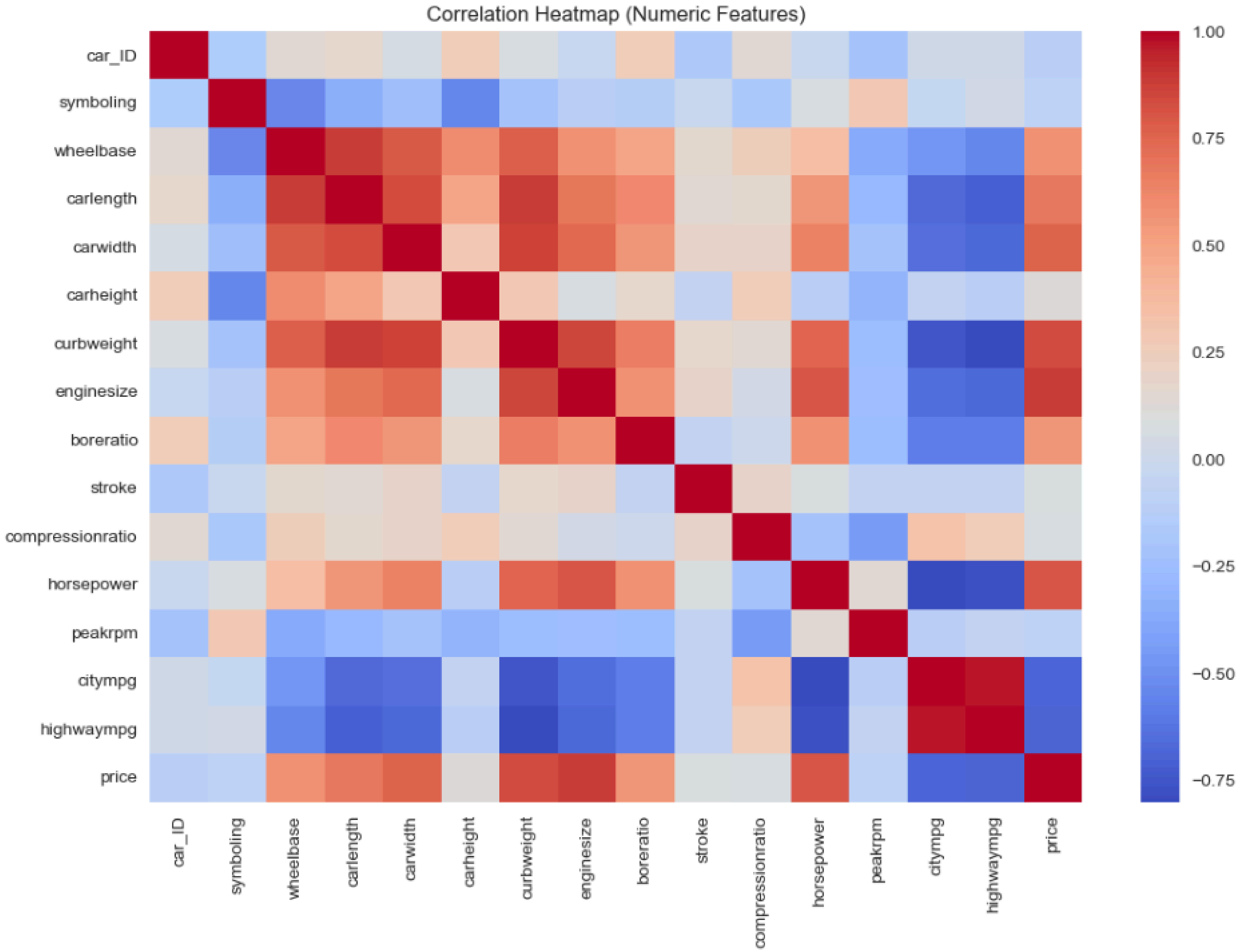
# NOTEBOOK 02: EXPLORATORY DATA ANALYSIS

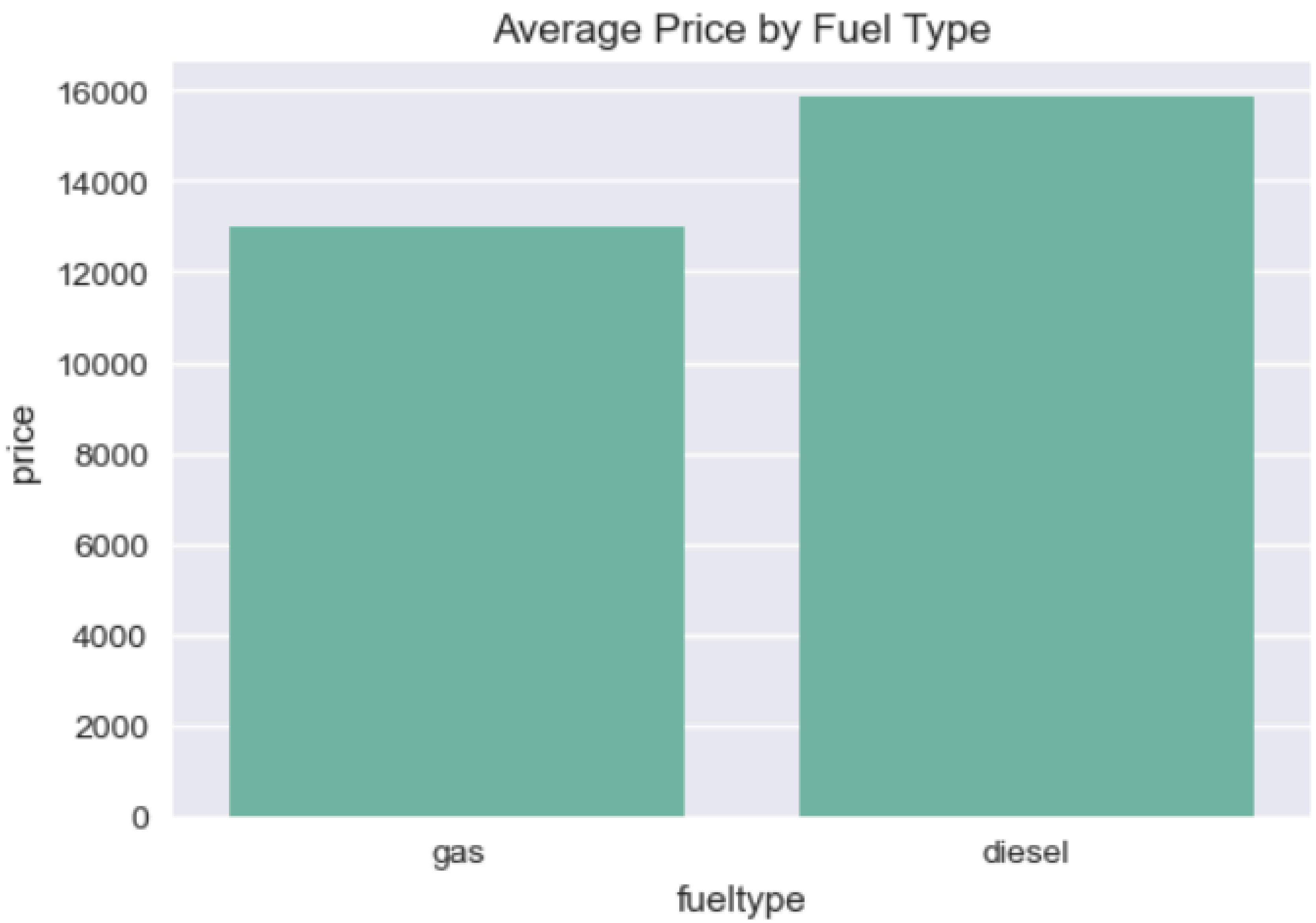


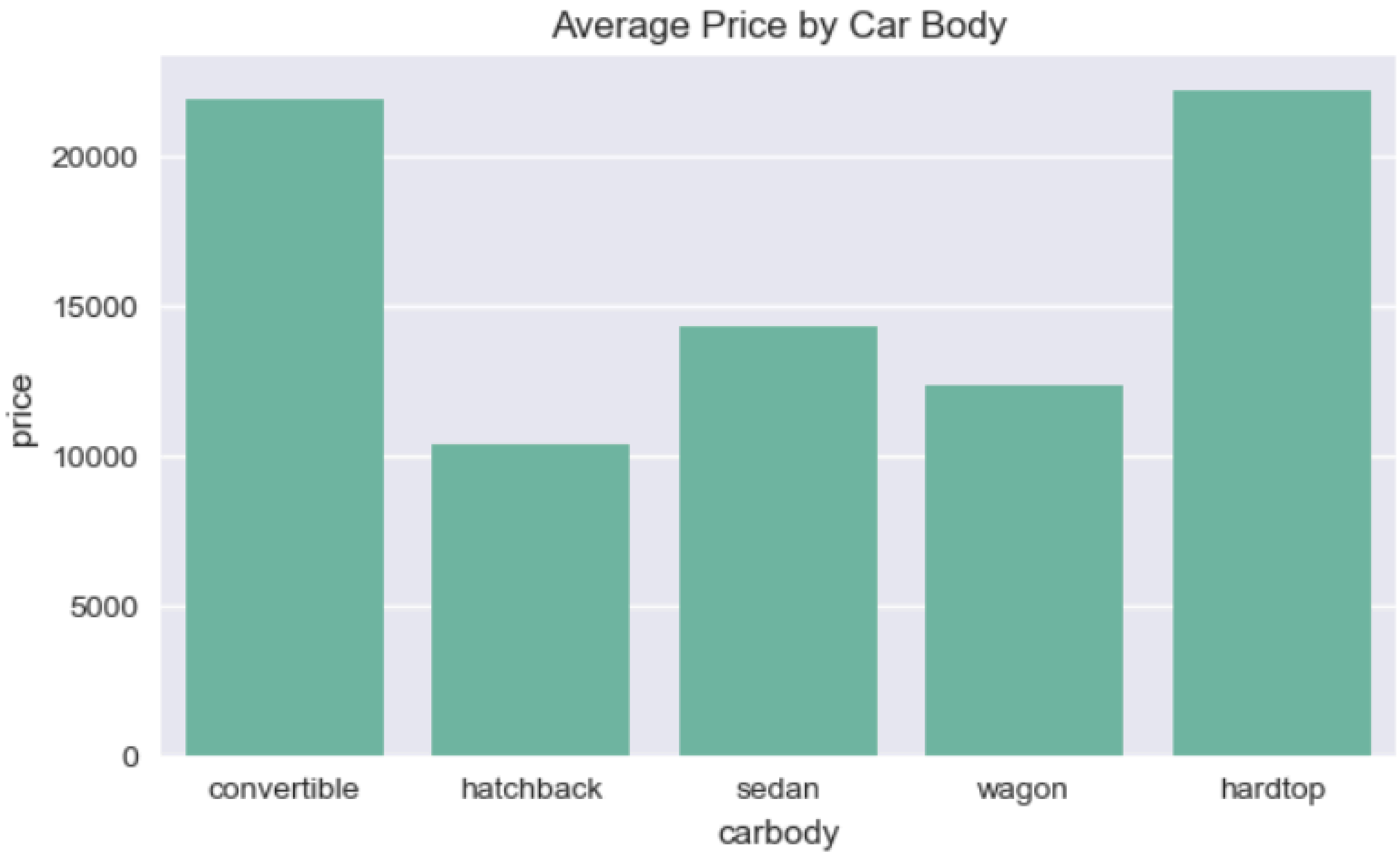
- Checked distribution of price
- Identified outliers in price and key numeric features
- Reviewed correlations between numeric features
- Compared price across categorical features (brand, body type, fuel type, drivewheel)
- Confirmed no leakage
- Saved cleaned dataset to processed folder




Correlation Heatmap (Numeric Features)
car_ID
symboling
wheelbase
carlength
carwidth
carheight
curbweight
enginesize
boreratio
stroke
compressionratio
horsepower
peakrpm
citympg
highwaympg
price
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75

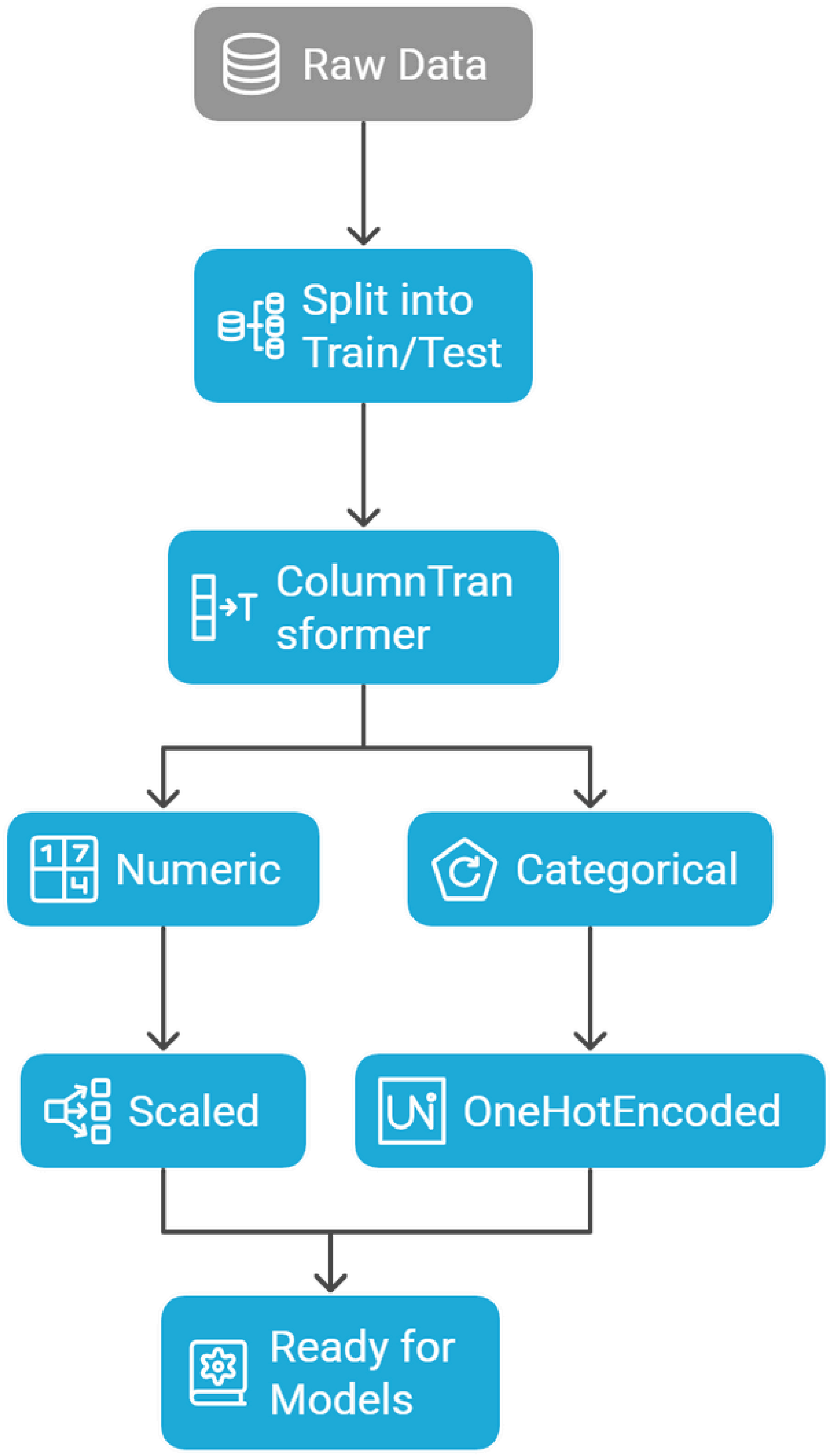

Average Price by Fuel Type
16000
14000
12000
10000
8000
6000
4000
2000
0
price
gas
diesel
fueltype


Average Price by Car Body
20000
15000
10000
5000
0
price
convertible
hatchback
sedan
wagon
hardtop
carbody

# DATA PREPROCESSING



- Split data: 80% training, 20% testing
- Separated numeric and categorical features
- Scaled numeric columns using StandardScaler
- Encoded categorical columns using OneHotEncoder
- Combined both using ColumnTransformer
- Ensured transformations were applied consistently to train and test sets



# NOTEBOOK 03: BASELINE MODELS



Tested 5 regression models:

- Linear Regression
- Ridge Regression
- Lasso Regression
- Decision Tree Regressor
- Random Forest Regressor

Used consistent preprocessing pipeline for all

Evaluated using:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- $R^2$ Score

| | Model | MAE | RMSE | $R^2$ |
|---|---|---|---|---|
| 0 | Random Forest | 1380.47 | 1938.93 | 0.9524 |
| 1 | Ridge Regression | 1823.77 | 2762.44 | 0.9033 |
| 2 | Lasso Regression | 1913.83 | 2845.79 | 0.8974 |
| 3 | Decision Tree | 2098.49 | 3427.12 | 0.8512 |
| 4 | Linear Regression | 3152.93 | 4961.85 | 0.6881 |

# NOTEBOOK 04: OPTIMIZED MODELS (GridSearchCV)



-Used GridSearchCV with 5-fold cross-validation
-Tuned key hyperparameters for advanced models:

- Random Forest Regressor
- Gradient Boosting Regressor
- XGBoost Regressor
- SVR (Support Vector Regressor)
- KNN Regressor

-Compared tuned models using MAE, RMSE, and $R^2$
-Selected best performing model for final evaluation

Optimized models (sorted):

| | Model | MAE | RMSE | $R^2$ |
|---|---|---|---|---|
| 0 | RandomForest | 1354.08 | 1895.05 | 0.9545 |
| 1 | GradientBoosting | 1822.26 | 2360.91 | 0.9294 |
| 2 | KNN | 2036.45 | 3244.23 | 0.8667 |
| 3 | SVR | 3066.88 | 5593.03 | 0.6037 |

# FINAL MODEL PERFORMANCE



- Best model: Optimized Random Forest Regressor
- $R^2$: 0.9545 (best overall score)
- RMSE: ≈ 1939
- MAE: ≈ 1380
- Reason for success:

1. Captures non-linear relations
2. Handles feature interactions
3. Reduces overfitting through averaging

Optimized models (sorted):

| | Model | MAE | RMSE | $R^2$ |
|---|---|---|---|---|
| 0 | RandomForest | 1354.08 | 1895.05 | 0.9545 |
| 1 | GradientBoosting | 1822.26 | 2360.91 | 0.9294 |
| 2 | KNN | 2036.45 | 3244.23 | 0.8667 |
| 3 | SVR | 3066.88 | 5593.03 | 0.6037 |

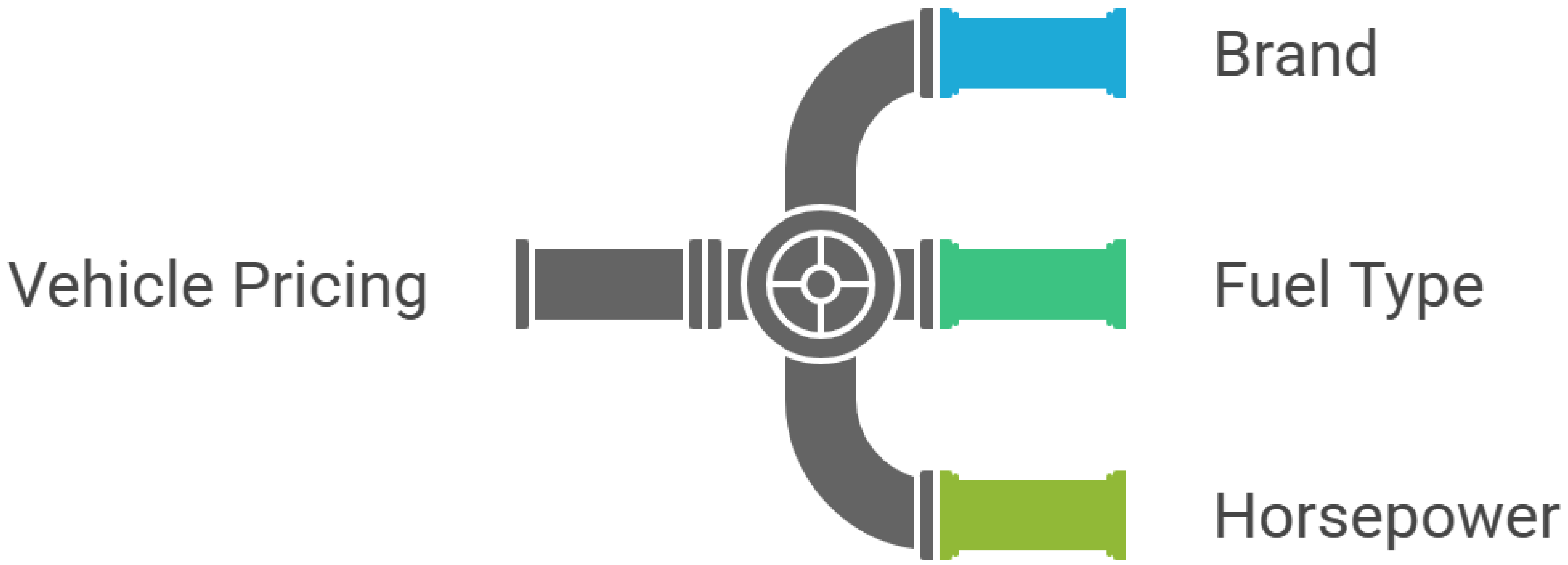
# CHALLENGES & LEARNINGS



- Small dataset (205 rows) → limited training information
- Some categorical columns had typos and inconsistent names
- Price distribution was skewed → outlier impact
- Needed to combine numeric and categorical features properly
- Training and GridSearch took time → handled with smaller parameter ranges
- Learned full workflow: EDA → Preprocessing → Modeling → Optimization

# FUTURE WORK






- Add more recent or larger car price datasets
- Integrate web scraping to get live market data (e.g. from car listing sites)
- Apply advanced models (CatBoost, LightGBM, or deep learning regression)
- Build a Streamlit web app to predict price based on car features
- Add feature importance visualizations (SHAP values) for explainability
- Deploy model on GitHub or cloud service for public testing

# CONCLUSION



- Built an end-to-end car price prediction system
- Structured the project professionally (raw → processed → notebooks → models)
- Explored data, cleaned it, and understood key patterns
- Compared multiple regression models
- Optimized advanced models using GridSearchCV
- Achieved strong performance with Random Forest ($R^2 \approx 0.95$)
- Ready for deployment or future extension

Optimized models (sorted):

| | Model | MAE | RMSE | $R^2$ |
|---|---|---|---|---|
| 0 | RandomForest | 1354.08 | 1895.05 | 0.9545 |
| 1 | GradientBoosting | 1822.26 | 2360.91 | 0.9294 |
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Baseline models (sorted):

| | Model | MAE | RMSE | $R^2$ |
|---|---|---|---|---|
| 0 | Random Forest | 1380.47 | 1938.93 | 0.9524 |
| 1 | Ridge Regression | 1823.77 | 2762.44 | 0.9033 |
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# Thank you